

"Apni Sone Ki Chidiya Ko Phir Se Udaan Do" ■■

A full-stack, AI-powered multilingual voice and chat welfare assistant connecting Indian citizens with verified government schemes, scholarships, and cyber safety tools.

Detailed project guides are available in the repository:

- **■ [PROJECT_DOCUMENTATION.md](#)** — Full technical architecture guide, data flow, component hierarchy, security protocols, and installation guide.
- **■ [TEAM_ROLES_DIVISION.md](#)** — Complete 4-member role ownership matrix with individual technical deliverables and USPs.

Each team member spearheads a distinct technological pillar that powers AmritChidiya:		
Team Member	Engineering Role & Focus	Member USP (Unique Selling Proposition)
Member 1	Frontend UI/UX & Citizen Access Lead	Phygital Last-Mile Bridge & Citizen CSC Dossier Generator Bridges the digital divide by generating physical, printable CSC Scheme Dossiers (AC-CSC-XXXXXX) with NPCI/DBT bank checklists & VLE operator sign-off boxes for village Jan Seva Kendras, coupled with a responsive Next.js 16 glassmorphism UI & live audio Talk Mode.
Member 2	Backend API & Speech Processing Lead	Low-Latency Vernacular Audio Pipeline with Indian Accent DSP Engineered an end-to-end voice pipeline using Groq Whisper Large v3 with phonetic domain prompting for Indian vernacular speech, background silence/hallucination suppression, and high-fidelity Microsoft Edge-TTS neural streaming in 5 Indian languages.
Member 3	AI Agent, Scheme Logic & Cyber Safety Lead	Anti-Hallucination Welfare Match Engine & Domain Cyber Shield Built the LangGraph dual-phase state machine ensuring 0% hallucinated welfare schemes, combined with the Official Domain Shield & Scam Checker that cryptographically verifies .gov.in / .nic.in URLs and shields citizens from WhatsApp fraud.
Member 4	Database, Auth Security & DevOps Lead	Zero-Trust Hybrid Cloud Persistence with Bcrypt Salted Auth Architected an enterprise-grade dual database layer (Supabase Cloud PostgreSQL + auto SQLite failover replica), hardened by salted Bcrypt (12 rounds) , signed 24-hour JWT tokens , SlowAPI rate limiting, and production cross-origin deployment.

■ Core Feature Highlights

- **■ Multilingual Conversational AI:** Native conversational support across **5 languages** (Hindi, Hinglish, English, Marathi, Tamil).
- **■ Hands-Free Talk Mode:** Full-duplex conversational voice mode powered by Groq Whisper Large v3 and Edge-TTS neural voices.
- **■ Yojana Eligibility Calculator:** Interactive multi-criteria matrix dynamically evaluating State, Category, Income, and Profession to estimate annual financial benefits.
- **■ Cyber Shield & Fake Scheme Checker:** Instant link verification tool protecting citizens from fraudulent WhatsApp schemes by verifying [.gov.in](#) and [.nic.in](#) domains with 1930 Cyber Helpline integration.
- **■ Printable CSC Dossier:** One-click printable PDF/dossier for village Common Service Centers (Jan Seva Kendra) with mandatory document checklists (Aadhaar, DBT/NPCI bank passbook).
- **■ Direct Official Portal Links:** Verified links to official central and state portals ([scholarships.gov.in](#), [scholarship.up.gov.in](#), [pmkisan.gov.in](#), [myscheme.gov.in](#)).
- **■ Hardened Security & Privacy:** 12-round Bcrypt password hashing, HS256 JWT authorization, strict IDOR ownership checks, SlowAPI rate-limiting, and ephemeral audio cleanup.

Enterprise Security & Privacy Matrix

AmritChidiya enforces defense-in-depth across 7 security and privacy layers:

Security Pillar	Code Implementation	Threat / Risk Mitigated
1. Credential Encryption	<code>bcrypt.gensalt(rounds=12)</code> in <code>database.py</code> with auto-migration from legacy SHA-256.	Rainbow table attacks, GPU brute-forcing, password database leaks.
2. Session & IDOR Defense	HS256 JWT tokens (24h expiry) + strict user-ownership validation on <code>/chats/{user_id}</code> .	Insecure Direct Object References, token spoofing, unauthorized chat access.
3. HTTP Security Headers	<code>SecurityHeadersMiddleware</code> enforcing <code>nosniff</code> , <code>DENY</code> clickjacking, XSS filter, strict referrer.	MIME sniffing, framing/clickjacking attacks, cross-origin referer data leaks.
4. DoS & Abuse Throttling	SlowAPI rate limits on <code>/signup</code> (5/min), <code>/login</code> (10/min), <code>/chat</code> (30/min), STT (20/min).	Bot registration floods, brute-force dictionary attacks, LLM API quota drain.
5. Voice Data Privacy	<code>tempfile.NamedTemporaryFile</code> + guaranteed <code>os.unlink()</code> cleanup in <code>finally</code> blocks.	Persistent citizen voice recordings on server disks, audio privacy leakage.
6. AI Anti-Hallucination	Canonical <code>SCHEME_URL_MAP</code> + <code>myscheme.gov.in</code> fallback; Whisper silence hallucination filter.	Fake welfare portals, phishing links, ghost queries generated on ambient room noise.
7. Citizen Anti-Fraud Shield	<code>ScamShieldModal</code> (4 Golden Rules + 1930 Cyber Helpline); guidance vs direct apply disclaimers.	WhatsApp advance fee scams, Aadhaar OTP theft, deceptive dark patterns.

■ ■ Technology Stack

- **Frontend:** Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, Lucide Icons, ReactMarkdown
- **Backend:** FastAPI (Python 3.14), LangGraph, LangChain, Groq LLM ([qwen/qwen3.8-27b](#))
- **Speech & Audio:** Groq Whisper Large v3 (STT), Microsoft Edge-TTS (Neural Voice Synthesis)

- **Database & Auth:** Supabase Cloud (PostgreSQL) + Local SQLite3 Fallback, Bcrypt, Python-JOSE (JWT)
- **Security & Infrastructure:** SlowAPI (Rate Limiting), Starlette Security Headers, CORS Origin Regex

■ Quick Start Guide

1. Backend Server

```
cd backend
python -m venv .venv
source .venv/bin/activate # On Windows: .venv\Scripts\activate
pip install -r requirements.txt
python main.py
```

Backend runs on <http://127.0.0.1:8000>.

2. Frontend Application

```
cd frontend
npm install
npm run dev
```

Frontend runs on <http://localhost:3000>.